

ELE214 - Electronics Laboratory I Preliminary Work 3

1 Questions

1. Explain how you can test a BJT whether it is defective or not. Also propose a method to determine the type of the BJT (NPN or PNP).
2. Draw the input characteristics curve ($I_B - V_{BE}$) for a typical transistor taking three different V_{CE} values.
3. Draw the output characteristics curve ($I_C - V_{CE}$) for a typical transistor taking three different base currents. Indicate the cut-off, active, and saturation regions on the curve.
4. Read the experimental section carefully to ensure the purpose of each step.

Experimental Work (Simulation of 2-3)

1. Test the BC177 transistor and determine its type.

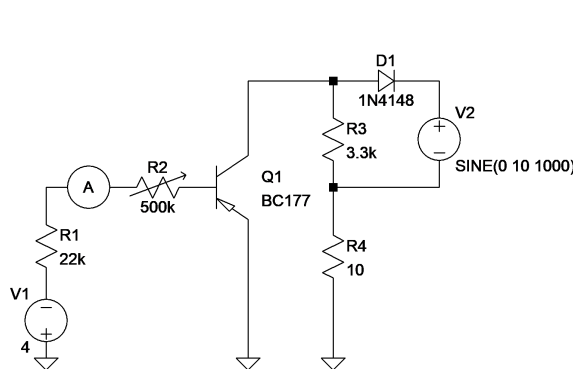


Figure 1

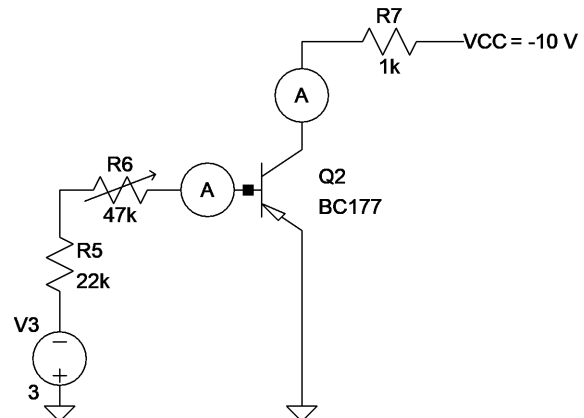


Figure 2

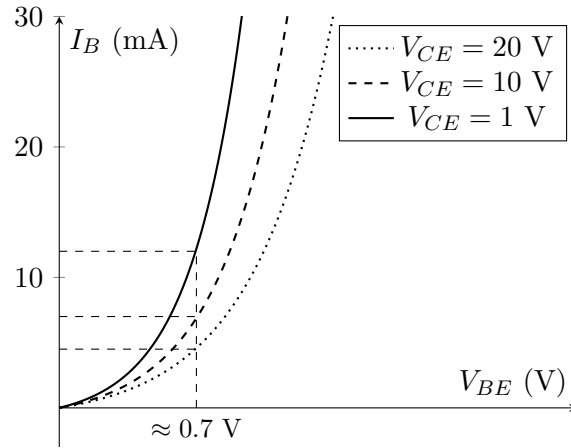
2. Set up the circuit in *Figure 1* setting $V_s = 10 \sin(2\pi(1000)t)$. Observe the output characteristics on the oscilloscope for three different values of base current. Plot the results.
3. Set up the circuit in *Figure 2*. For three positions of the potentiometer (minimum, medium and maximum resistance), measure collector currents for the base currents in the previous part. Also record the base resistances for each case.

2 Answers

1. We can test the functionality of a BJT transistor by means of a multimeter. In order for the BJT to be healthy, there shouldn't be conduction between the collector and the emitter if the back-to-back diodes analogy is considered. If there is no conduction between any terminals, then the BJT acts as an open-circuit, which we don't intend.

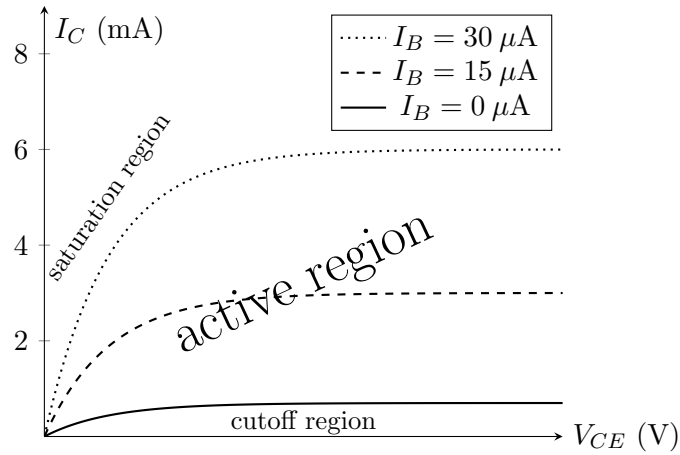
After this step, to determine whether it is an NPN or PNP transistor, put the red probe into the base and the black probe into any other terminal. If there's a voltage drop, then it is an NPN transistor. On the contrary, if the *OL* reading is obtained, it is a PNP transistor.

2. As V_{CE} increases, the graph is more likely to shift downwards due to the Early effect.



Graph 1: Input characteristics curve for a BJT

3. The graph is given.



Graph 2: Output characteristics curve for a BJT

Simulation

2. The graph is given below.

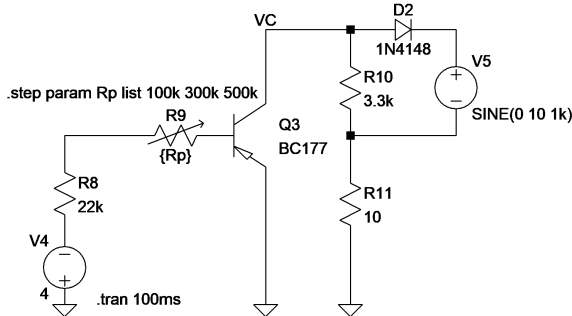
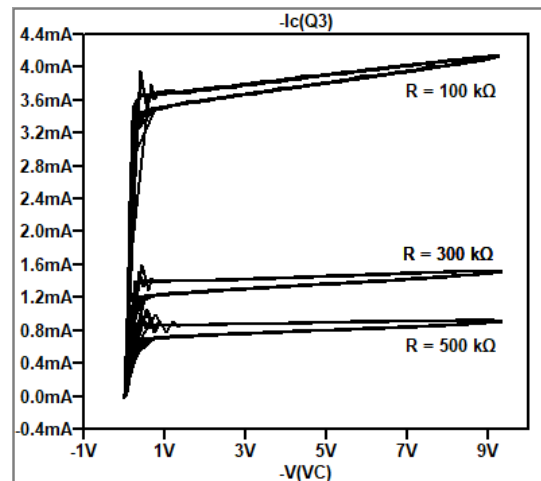


Figure 3: Figure 1 adjusted



Graph 3: Output characteristics curve for Figure 3

$$I_C = -4.14 \text{ mA for } R = 100 \text{ k}\Omega,$$

$$I_C = -1.51 \text{ mA for } R = 300 \text{ k}\Omega,$$

$$I_C = -0.91 \text{ mA for } R = 500 \text{ k}\Omega.$$

3. The graph is given below.

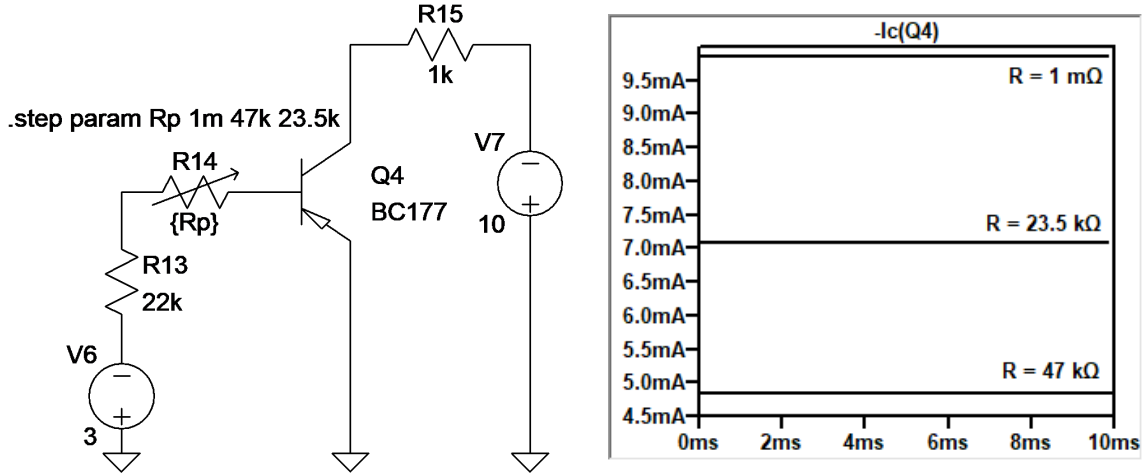


Figure 4: Figure 2 adjusted

Graph 4: Current values for Figure 4

$$I_C = -9.85 \text{ mA for } R = 1 \text{ m}\Omega \approx 0 \Omega,$$

$$I_C = -7.10 \text{ mA for } R = 23.5 \text{ k}\Omega,$$

$$I_C = -4.86 \text{ mA for } R = 47 \text{ k}\Omega.$$

Base resistances:

$$R_B = 0 + 22 \text{ k} = 22 \text{ k}\Omega \text{ for } I_C = -9.85 \text{ mA},$$

$$R_B = 23.5 + 22 \text{ k} = 45.5 \text{ k}\Omega \text{ for } I_C = -7.10 \text{ mA},$$

$$R_B = 47 + 22 \text{ k} = 69 \text{ k}\Omega \text{ for } I_C = -4.86 \text{ mA}.$$